RingMo-Galaxy: A Remote Sensing Distributed Foundation Model for Diverse Downstream Tasks

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***Abstract*— Remote sensing lightweight foundation models have successfully achieved online perception, providing real-time intelligent interpretation. However, their capabilities are restricted to inferences solely based on their respective observations and models, thus lacking a comprehensive understanding of large-scale remote sensing scenarios. To address this limitation, we propose RingMo-Galaxy, a remote sensing distributed foundation model based on generalized information mapping and interaction. RingMo-Galaxy can realize online collaborative perception across multiple platforms and diverse downstream tasks by mapping observations into a unified space and implementing a task-agnostic information interaction strategy. Specifically, we leverage the ground-based geometric prior of remote sensing oblique observations to change feature mapping from absolute to relative depth estimation, thereby enhancing the model's ability to extract generalized features across diverse heights and perspectives. In addition, we present a dual-branch information compression module to decouple high-frequency and low-frequency features, achieving feature-level compression while preserving critical task-agnostic details. To support our research, we collect a multitask simulation dataset named AirCo-MultiTasks, specifically designed for multi-unmanned aerial vehicle (UAV) collaborative observation. We also conduct extensive experiments, including 3-D object detection, instance segmentation, and trajectory prediction. The numerous results demonstrate that our proposed RingMo-Galaxy achieves state-of-the-art performance across various downstream tasks.**

***Index Terms*— Distributed foundation model, generalized feature mapping, information decoupling, multiplatform collaboration.**

I. Introduction

REMOTE sensing foundation models have been widely applied in remote sensing intelligent perception [1], [2], [3], relying on powerful generalization performance. Moreover, continuous advancements in computational resources and the rapid evolution of lightweight techniques, such as knowledge distillation and quantization, have provided the necessary conditions for online inference of remote sensing foundation models. For instance, RingMo-Lite [4] demonstrates the capability to perform online inference tasks, including object detection [5], [6] and semantic segmentation [7], [8], on a single-edge terminal.

Intuitively, remote sensing scenarios typically cover extensive areas and complex environment structures, posing challenges for lightweight edge models. For instance, single-platform observations are restricted by perspectives and resolutions, hindering the capture of comprehensive information on observed objects and leading to issues, such as occlusion and imaging disturbances. Therefore, it is necessary to consider the correlation and complementarity among multiview, multiresolution observations obtained from various platforms. Unfortunately, current remote sensing foundation models merely rely on single-platform observations to perform online perception, failing to fully utilize the advantages of multiplatform collaborative observation and computational resources. To this end, it is worthwhile to develop a distributed foundation model that can support multiplatform online collaborative perception, as illustrated in Fig. 1. This model can not only achieve independent online perception based on each platform's resources but also further investigate interplatform relationships for comprehensive collaborative analysis, significantly enhancing the performance of multiplatform, multitask online interpretation in complex and large-scale remote sensing scenarios.

However, there are two significant challenges in constructing an online collaborative perception model for multiple remote sensing platforms. The first challenge lies in the recognition inconsistency across different platforms. Significant variations in altitudes and perspectives among various platforms lead to discrepancies in the observations of identical scenes, as illustrated in the top of Fig. 1. Consequently, noticeable gaps emerge among multiplatform information, posing challenges for subsequent collaborative interactions. The second challenge is the information compression loss due to interplatform communication constraints [9], [10]. Complex environments and long-distance transmission limit the bandwidth available for massive information exchanges. Therefore, it is inevitable to reduce information transmission during collaborative interactions, such as through spatially sparse feature

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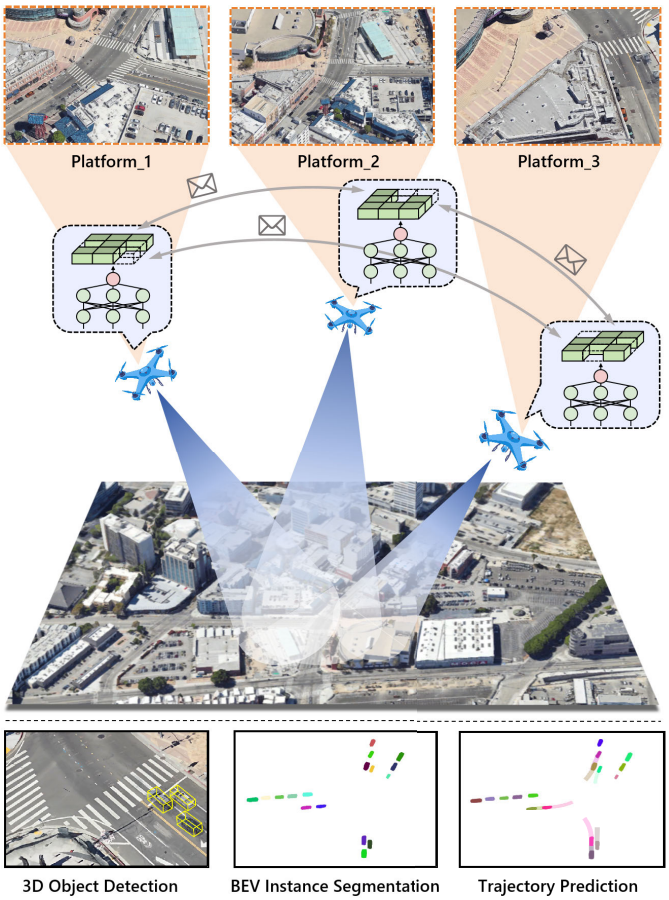


Fig. 1. Illustration of the collaborative perception among multiple remote sensing platforms. In this scene, several platforms observe the same area from different angles and heights. Through interplatform collaboration, they achieve a more comprehensive understanding of the scene, enhancing the performance of various downstream tasks.

maps [11]. However, such compression, primarily driven by the bounding box confidence and designed for detection tasks, may result in incomplete representations, making it insufficient for tasks that require rich feature details, such as segmentation and prediction. Consequently, this design can interfere with the local observations of other platforms, ultimately degrading the performance of multiplatform collaborative perception.

To address these challenges, this article introduces RingMo-Galaxy, a remote sensing distributed foundation model based on the principle of "physical separation and logical integration." The term "Galaxy" signifies a constellation of online foundation models, such as RingMo-Lite [4]. RingMo-Galaxy facilitates cooperation among these models by focusing on two key aspects: generalized information extraction from distributed observation platforms and selective collaborative interactions for comprehensive information integration. Specifically, a generalized BEV generalization (GBG) module is presented to achieve more generalized and precise bird's eye view (BEV) representations. The GBG module derives a more accurate depth of observed objects through both geometric priors from oblique observations and a relative estimation manner, which is essential for subsequent BEV representations in the unified space. The oblique observations provide a depth upper bound for each line of sight, and the adopted relative depth estimation avoids the traditional fixed depth interval settings, facilitating a GBG across platforms with varying altitudes and perspectives. In addition, a high–low-frequency decoupled collaboration (HLFDC) module is proposed as a task-agnostic manner to facilitate efficient interplatform interactions and fusion across various tasks. Specifically, the HLFDC module leverages distinct attention mechanisms to decouple and compress high-frequency and low-frequency features from each platform's BEV representations, ensuring low-cost transmission while preserving spatial continuity and retaining critical information. Then, transmitted collaborative features are restored to the same feature dimension as the local features and undergo pixel-level weighted fusion to construct a more comprehensive scene perception. Finally, a simulated dataset for multi-unmanned aerial vehicle (UAV) collaborative observation, named AirCo-MultiTasks, is collected to validate the effectiveness of our proposed RingMo-Galaxy. The dataset consists of five UAVs at different flight heights and perspectives and includes 160 scenes and 40k images, supporting multitasks such as detection, segmentation, and prediction. Experimental validation demonstrates that our model achieves state-of-the-art performance on various downstream tasks, while requiring only a 53.1% transmission ratio.

The main contributions of this article can be summarized as follows.

1) This article proposes RingMo-Galaxy, a remote sensing distributed foundation model that simultaneously achieves independent online perception on individual terminals as well as collaborative online perception across multiple platforms.
2) A GBG module is introduced to transform the significantly divergent 2-D observations from various altitudes and perspectives into generalized BEV representations within a unified feature space.
3) An HLFDC module is proposed for efficient collaborative information interactions and integration. This is accomplished by decoupling and compressing the high-frequency and low-frequency components of collaborative features, followed by pixel-level fusion of these features with local ones.
4) To validate the effectiveness of our proposed RingMo-Galaxy framework, we introduce a large-scale multitask multi-UAV collaborative perception dataset, AirCo-MultiTasks. In this dataset, RingMo-Galaxy achieves state-of-the-art performance in downstream tasks, such as detection, segmentation, and prediction, while maintaining relatively low transmission costs.

II. Related Works

A. Remote Sensing Foundation Model

The utilization of foundation models has become a standard practice in computer vision, offering robust baselines and improving performance across various tasks. For instance, vision transformer (ViT) [12] addresses the limitations of traditional convolutional neural network (CNN) by employing the Transformer architecture [13]. In addition, masked autoencoders (MAE) [14] enhances the model's robustness and representation capabilities through a transformer-based

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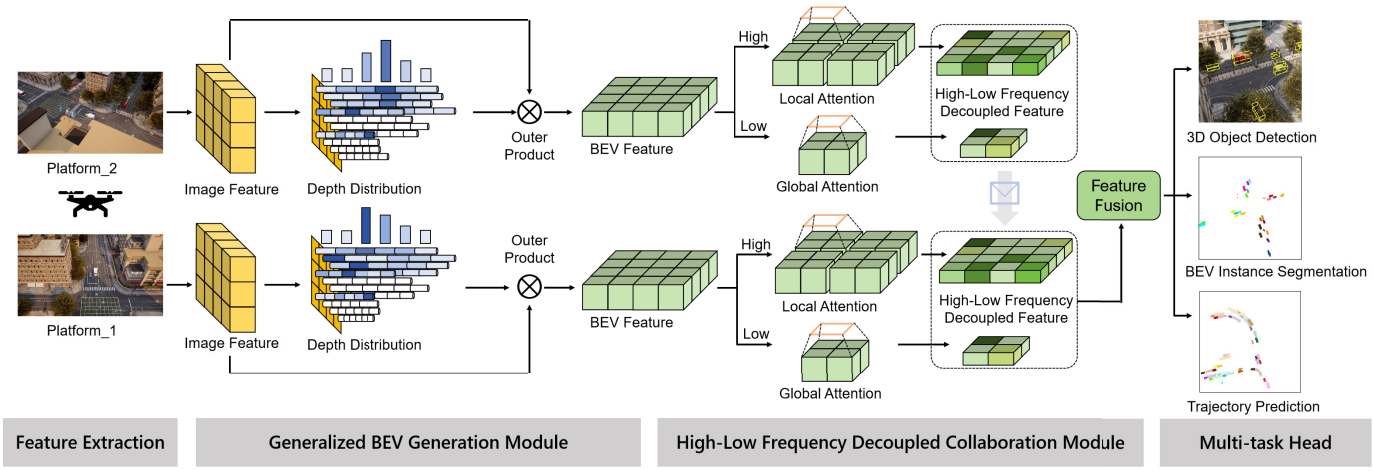


Fig. 2. Overall architecture of our proposed RingMo-Galaxy framework. For clarity, we just present the collaboration between the two platforms.

masked image reconstruction approach, while SimMIM [15] achieves efficient image modeling with a simplified design.

Intelligent remote sensing perception has also greatly benefited from the widespread adoption of foundation models, leading to the development of numerous advanced models. RingMo [16], the first generative self-supervised foundation model framework in the field of remote sensing, employs masked image modeling [17] to obtain general feature representations, improving the accuracy of various remote sensing perception tasks. Billionmodel [18] introduces the largest billion-scale foundation model in remote sensing, pushing the boundaries of model size exploration in this domain. RingMo-SAM [19] processes both optical and SAR remote sensing data [20], [21], [22], [23], [24], [25], enabling zero-shot semantic segmentation and object category identification with solid generalization capabilities. In addition, several studies have focused on parameter-efficient fine-tuning of remote sensing foundation models to enhance task performance [26], [27].

Despite their advantages, foundation models face challenges in terms of flexibility and efficiency due to their high computational and storage demands, making them difficult to deploy on edge servers or terminals. This limitation hinders their ability to support on-orbit perception in practical remote sensing applications. To address this issue, RingMo-Lite [4] is introduced to alleviate the high computational requirements, facilitating the execution of remote sensing foundation models on edge devices while maintaining high accuracy and versatility across multiple tasks.

B. Collaborative Perception

Collaborative perception integrates information from multiple observation platforms [28], [29], [30] to enhance holistic perception, thereby improving online intelligent interpretation. Existing methods can generally be grouped into three categories: early collaboration based on raw data, late collaboration based on final results, and intermediate collaboration focused on feature sharing [31]. Early collaboration [32], which aggregates raw data from various platforms, typically achieves higher performance but incurs significant transmission costs, rendering it impractical under bandwidth limitations. Late collaboration [33], which involves sharing prediction outcomes, reduces transmission costs but can result in considerable information loss and potential conflicts in predictions. Intermediate collaboration [34], [35], which shares intermediate-layer features, offers a compromise by preserving semantic information while controlling transmission overhead. This paradigm is more suitable for real-time collaborative interpretation and has also been adopted in this study.

For the intermediate collaboration, several studies have demonstrated its effectiveness in autonomous driving. For instance, Who2com [34] implements a handshake mechanism to select collaborative partners, while When2com [35] determines the optimal time to initiate collaboration. Where2com [11] uses the detection head to guide interactions in specific regions. CoCa3D [36] focuses on collaborative depth estimation by integrating single-view depth probability with multiview consistency. Heterogeneous alliance (HEAL) [37] proposes a heterogeneous collaborative framework that allows the incorporation of new agents. CodeFilling [38] introduces a codebook-based representation and an information-filling-driven selection process to balance perception and communication.

Collaborative perception also plays a significant role in remote sensing scenarios [39]. MDCNet [40] proposes a multiplatform distributed collaborative inference network for object detection, overcoming the occlusion of single-platform observations and improving overall detection precision through a sparsely spatial-level strategy for interplatform selective interaction. DCP-Net [41] designs an agent-level collaborative interaction strategy, considering feature deviations between platforms to achieve collaborative semantic segmentation. DCNNet [42] presents a terminal-cloud collaborative, progressive inference mechanism for image classification, employing a distributed self-distillation paradigm to refine deep features and facilitate efficient knowledge transfer between terminals and the cloud network. However, these collaborative perception approaches in remote sensing are relatively limited to specific tasks and fail to account for the significant differences in observations among platforms. Our proposed RingMo-Galaxy is committed to mitigating these insufficiencies.

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III. Methods

This section presents RingMo-Galaxy, a well-designed collaborative perception framework for remote sensing platforms, enabling multiple platforms to share visual information and thereby promoting more holistic perception. As depicted in Fig. 2, RingMo-Galaxy comprises four components (i.e., feature extraction, GBG module, HLFDC module, and supports for multiple downstream tasks) to implement the entire collaboration process. We briefly introduce these components in the overview, with a detailed exploration of the innovative designs of the GBG and HLFDC modules in Sections III and IV, respectively.

A. Overview

The RingMo-Galaxy framework is composed of the following components.

1) Feature Extraction: Initially, each platform captures and processes its observations using a shared encoder to extract semantic information. We utilize the Swin-Tiny [43] as the encoder for each drone, which is a lightweight foundational model pretrained on ImageNet-1k. This choice is due to Swin-Tiny's low inference latency and optimized memory usage within the family of transformer-based backbones.

2) GBG Module: For collaboration among platforms, image features need to be transformed from each pixel coordinate to BEV representations in the world coordinates. Based on the encoded image features, the GBG module provides generalized depth estimation for various platforms at different angles and heights by leveraging the geometric prior. Then, image features and their corresponding estimated depths are used to generate frustum features, which are then converted into voxel features with the help of the camera's intrinsic and extrinsic parameters. Ultimately, z-axis sum-pooling operation and height dimensionality reduction are applied to the voxel features to derive BEV representations, which are convenient for later collaboration.

3) HLFDC Module: The HLFDC Module is designed to balance enhanced perception with transmission costs during interplatform collaboration. Given the characteristics of remote sensing images, such as small and widely distributed objects, and a large proportion of background, this module achieves efficient information compression by decoupling high-frequency and low-frequency components. High-frequency components, which capture fine details, are meticulously preserved, while low-frequency components, representing the global structure, are spatially compressed. Notably, both components undergo channelwise compression, reducing their dimensions to half of the original features. After collaborative interactions, local features and transmitted compressed collaborative features are fused via pixelwise weighting, thus enhancing subsequent downstream tasks.

4) Supports for Multiple Downstream Tasks: The collaboration among multiple platforms aims to enrich their respective perceptual information and is not limited to a specific task. Consequently, RingMo-Galaxy supports various downstream tasks, including 3-D object detection, BEV instance segmentation, and multiobject trajectory prediction. Specifically, the fused features generated by the HLFDC module are fed into downstream task heads to produce enhanced perception results.

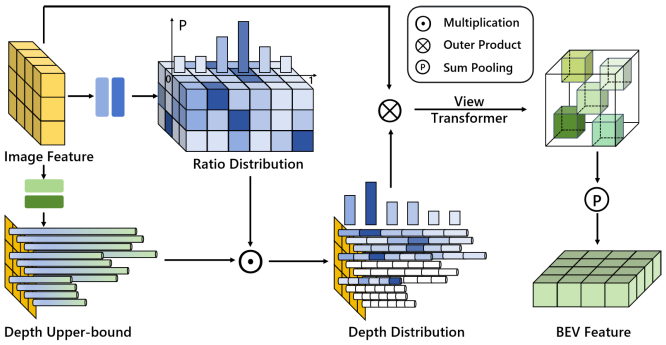


Fig. 3. Operation flow of the GBG module. This module introduces generalized feature mapping through relative depth estimation, which enhances the accuracy of BEV representation.

B. GBG Module

Various remote sensing platforms can cause significant discrepancies in the observations of identical objects due to different viewing angles and altitudes. Consequently, it is essential to convert multiview observations into a unified spatial coordinate before information fusion. To this end, we transform the 2-D image features to the BEV coordinate system. The BEV system offers distinct advantages, particularly in terms of scale and spatial consistency, which can effectively mitigate gaps among multiview observations. Depth estimation serves as a critical component in generating BEV representations. However, the vast range of aerial observations, coupled with various estimation intervals, presents challenges for the prevalent lift-splat-shot (LSS) method [44] in approximating the depth of each pixel. To address this issue, we propose an innovative GBG module, as illustrated in Fig. 3. This module leverages both the geometric priors provided by inclined observations and a relative depth estimation manner. The procedural methodology of the proposed GBG module is outlined as follows.

1) Derivation of Geometric Constraint: In this section, we utilize the camera's intrinsic and extrinsic parameters to calculate the corresponding depth upper bound for each position in the observation.

According to the pinhole camera model, the transformation between a point (x, y, z) in the world coordinate system, and its projection (u, v) in the pixel coordinate system can be described by the following equation:

$$\mathbf{d}\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} = \begin{bmatrix} \mathbf{K} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{R} & \mathbf{T} \\ \mathbf{0}^{\mathrm{T}} & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}. \tag{1}$$

Here, $\mathbf{K} \in \mathbb{R}^{3\times3}$ denotes the intrinsic matrix of the camera, and the extrinsics are represented by a rotation matrix $\mathbf{R} \in \mathbb{R}^{3\times3}$ and a translation vector $\mathbf{T} \in \mathbb{R}^{3\times1}$. The symbol d represents the exact pixel depth in the camera coordinate system. Subsequently, reverse the above process to derive

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the representations of the world coordinate from the pixel coordinate

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \mathbf{R}^{-1}\left(\mathbf{K}^{-1}\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} d - \mathbf{T}\right). \tag{2}$$

For notational brevity, we denote $\mathbf{R}^{-1}\mathbf{K}^{-1}$ as the matrix $\mathbf{M} = (m_{ij}) \in \mathbb{R}^{3\times 3}$ and $\mathbf{R}^{-1}(-\mathbf{T})$ as the vector $\mathbf{N} = (n_i) \in \mathbb{R}^{3\times 1}$. Hence, (2) is modified as follows:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \mathbf{M}\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} d + \mathbf{N} = \begin{bmatrix} m_{11} & m_{12} & m_{13} \\ m_{21} & m_{22} & m_{23} \\ m_{31} & m_{32} & m_{33} \end{bmatrix}\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} d + \begin{bmatrix} n_1 \\ n_2 \\ n_3 \end{bmatrix}. \tag{3}$$

In reference to the world coordinate system, the ground plane is located in the negative direction of the drone's y-axis. By setting $y = -H$ in (3), where H denotes the altitude of the drone, (3) can be reformulated as a linear equation in one variable

$$-H = \begin{bmatrix} m_{21} & m_{22} & m_{23} \end{bmatrix}\begin{bmatrix} u \\ v \\ 1 \end{bmatrix} d + n_2. \tag{4}$$

Consequently, the depth upper bound of each position in the camera coordinate can be derived as follows:

$$D_{(u,v)} = \frac{-H - n_2}{m_{21}u + m_{22}v + m_{23}}. \tag{5}$$

Following the acquisition of the depth upper bound for each pixel, we utilize this geometric constraint as a prior for optimizing depth estimation.

2) Relative Depth Estimation: In contrast to the previous absolute depth estimation, we introduce a relative depth estimation approach. This method calculates the ratio of each pixel's depth to its depth upper bound, replacing the global, fixed numerical depth estimation interval with a flexible relative depth estimation interval (0, 1) that is scaled by the depth upper bound for each region. Specifically, we employ a parametric network $\Phi_{\text{depth}}(\cdot)$ that generates a pixelwise categorical distribution of depth, represented as $d_{\text{pred}} = \Phi_{\text{depth}}(F_{(u,v)}) \cdot D_{(u,v)} \in \mathbb{R}^D$. Here, D signifies the number of discretized depth bins. We treat the depth estimation as a classification problem to capture the inherent depth estimation uncertainty and reduce the impact of erroneous depth estimates, as suggested in LSS [44].

3) BEV Generation: The following process transforms 2-D observations into BEV representations based on the pinhole camera model defined in 2. First, operate the outer product on the encoded features, $F_{\text{2-D}} \in \mathbb{R}^{W\times H\times C}$, and the corresponding estimated depth, $D_{\text{pred}} = \left[d_{\text{pred}_{(h,w)}}\right]_{(h\in H, w\in W)} \in \mathbb{R}^{W\times H\times D}$, to obtain the frustum features, which aim to recover the spatial information of the image. Then, project the frustum features into BEV pillars using the camera's intrinsics and extrinsics. This operation converts multiview observations into a unified world coordinate system. Finally, perform z-axis sum pooling at each spatial position $(x, y, :)$, collapsing the height dimension to generate the BEV features, $F_{\text{BEV}} \in \mathbb{R}^{X\times Y\times C}$. The obtained BEV representations serve for subsequent information exchange among different platforms.

C. HLFDC Module

Communication among remote sensing platforms is often restricted, necessitating the partial transmission of valuable information to achieve efficient collaborative perception. As indicated in the literature [45], [46], the high-frequency components of feature maps depict object details, while low-frequency components typically capture the global structure. Given that remote sensing images contain small, dense foreground objects alongside rich background information, we tend to preserve intricate details and compress global spatial information for lightweight transmission. Therefore, we propose a collaborative strategy based on high-low frequency decoupling, as outlined in Algorithm 1 and illustrated in Fig. 4. Notably, this design is task-agnostic, making it broadly applicable across various applications. The implementation is as follows:

1) High-Frequency Information Extraction: The high-frequency branch is utilized to capture local dependencies that reflect fine details. To this end, we implement local window self-attention with the nonoverlapping window partition.

The original BEV feature map is segmented into multiple local feature maps using nonoverlapping windows of size $M \times M$. Each local feature map is then reshaped into a feature sequence, $F_{\text{seq}} \in \mathbb{R}^{M^2\times C}$, which serves as the input to the multihead self-attention layer. Each self-attention head computes the query ($\mathbf{Q}$), key ($\mathbf{K}$), and value ($\mathbf{V}$) matrices through a linear transformation of F_{seq}

$$\mathbf{Q} = F_{\text{seq}}\mathbf{W}_q, \mathbf{K} = F_{\text{seq}}\mathbf{W}_k, \mathbf{V} = F_{\text{seq}}\mathbf{W}_v, \mathbf{Q}, \mathbf{K}, \mathbf{V} \in \mathbb{R}^{M^2\times D_h} \tag{6}$$

where $\mathbf{W}_q$, $\mathbf{W}_k$, and $\mathbf{W}_v \in \mathbb{R}^{C\times D_h}$ are learnable parameters, with D_h being the number of hidden dimensions for a head. The output of a self-attention head is then computed as a weighted sum over M^2 value vectors

$$\text{SA}_h(F_{\text{seq}}) = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{D_h}}\right)\mathbf{V} \in \mathbb{R}^{M^2\times D_h}. \tag{7}$$

For the multihead self-attention layer with N_h heads, the final output is obtained through a linear projection of the concatenated outputs from each self-attention head, expressed as follows:

$$\text{MSA}_h(F_{\text{Seq}}) = \text{concat}_{h\in[N_h]}[\text{SA}_h(F_{\text{Seq}})]\mathbf{W}_o \in \mathbb{R}^{M^2\times C'} \tag{8}$$

where $\mathbf{W}_o \in \mathbb{R}^{(N_h\times D_h)\times C'}$ is a learnable parameter. Herein, we set $C' = C/2$ and restructure the high-frequency feature sequences into their spatial format, which are then combined to form the complete high-frequency spatial feature $F_{\text{high}} \in \mathbb{R}^{X\times Y\times C'}$.

2) Low-Frequency Information Extraction: Several studies [47] have indicated the effectiveness of the global attention mechanism in capturing low-frequency information. However, due to the limited computational budget, it is expensive to directly perform global attention on original high-resolution feature maps. Inspired by Voigtman and Winefordner [48], we first initially utilize an average pooling operation to achieve dual functions of low-pass filtering and downsampling the feature maps. Considering the complementarity between high

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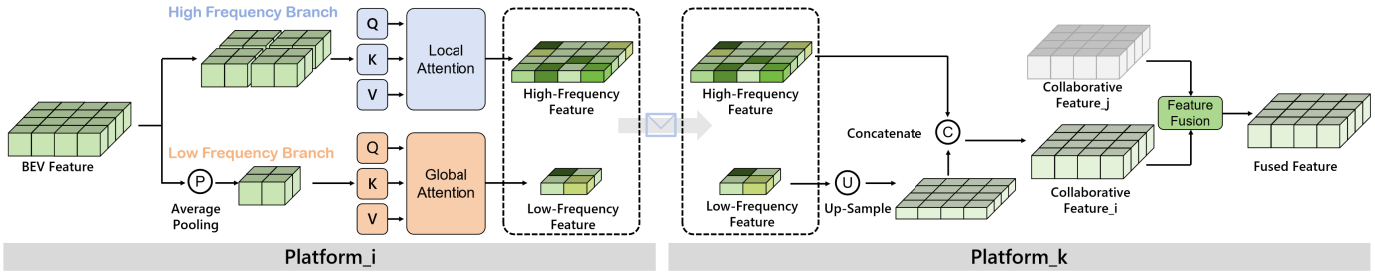


Fig. 4. Operation flow of the HLFDC module. This module compresses the original local features through two branches: low frequency and high frequency. In the high-frequency branch, the local attention is conducted to obtain high-frequency features reflecting objects' edges, shapes, and other details. These high-frequency features are then condensed into low-dimensional channels. Conversely, the low-frequency branch preserves the overall structural information through low-filtered average pooling and global attention. The high-frequency branch condenses spatial information into downsampled representations.

and low frequencies, the stride of the average pooling is set to match the size of the local window in the high-frequency branch.

Specifically, the original BEV feature map is average-pooled with a kernel and stride of size M to obtain the initial low-frequency signals, which are projected into N_h queries, keys, and values, $\mathbf{Q}, \mathbf{K}, \mathbf{V} \in \mathbb{R}^{XY/M^2 \times D_h}$, respectively. Next, global multihead self-attention is applied to capture the rich low-frequency feature sequences, which are subsequently reshaped to the low-frequency spatial features $F_{\text{low}} \in \mathbb{R}^{X/M \times Y/M \times C'}$. The process of multihead self-attention follows the same steps as described in (6)–(8).

3) Interplatform Collaborative Features Fusion: During the collaborative information exchange process, each platform transmits its high-low frequency decoupled features to the others to facilitate collaborative perception. To simplify implementation, this study focuses on a primary platform that receives information from other platforms. Since these features are generated in each platform's local coordinate system, it is necessary to convert them to the receiving one's coordinate system. Upon receiving these collaborative features, a geometric transformation G is applied to align them spatially with the local coordinate system. Subsequently, the platform integrates high-frequency and low-frequency components to reconstruct the original information across both the spatial and channel dimensions to ensure consistency in spatial scale and feature dimensionality. As the low-frequency features are average pooled, a series of transposed convolutions ConvTrans are utilized to upsample them to the original size, matching their shape with the high-frequency features. Consequently, the processed high-low frequency features are concatenated to generate the collaborative features F_{col} for subsequent fusion, which can be expressed as follows:

$$F_{\text{col}} = \text{concat}[\text{ConvTrans}(G(F_{\text{low}})); G(F_{\text{high}})] \in \mathbb{R}^{X \times Y \times C}. \tag{9}$$

Guided by local features F_{BEV}, the contribution weight W_j of $F_{\text{col,j}}$ toward constructing fused features F_{fused} is quantified by

$$W_j = \frac{\phi_{\text{conv}}\left([F_{\text{BEV}}; F_{\text{col,j}}]\right)}{\sum_{j=1}^{N} \phi_{\text{conv}}\left([F_{\text{BEV}}; F_{\text{col,j}}]\right)} \in \mathbb{R}^{X \times Y \times 1} \tag{10}$$

where ϕ_{conv} represents the multilayer convolutional operations. Moreover, a pixel-level weighted fusion is executed to generate the fused features for subsequent downstream tasks

$$F_{\text{fused}} = \sum_{j=1}^{N} W_j F_{\text{col,j}} \in \mathbb{R}^{X \times Y \times C}. \tag{11}$$

Algorithm 1 HLFDC Module

Basic Function Definitions:

High_Frequency_Extraction(F_{BEV}):
  Divide F_{BEV} into $M \times M$ non-overlapping windows;
  for each window **do**
    Apply local self-attention;
  end for
  Concatenate and reshape to form F_{high};
Low_Frequency_Extraction(F_{BEV}):
  Down-sample F_{BEV} by M;
  Apply global self-attention;
  Reshape to form F_{low};
Decoupled_Features_Recovery(F_{high}, F_{low}):
  Up-sample F_{low} via transposed convolution;
  Concatenate F_{low} and F_{high} to obtain F_{col};

Overall Operation Process:

Stage 1: High-Low Frequency Decoupling
  F_{high} = ***High_Frequency_Extraction***(F_{BEV});
  F_{low} = ***Low_Frequency_Extraction***(F_{BEV});
Stage 2: Information Transmission
  for each other platform **do**
    Receive and align decoupled features;
    F_{col} = ***Decoupled_Features_Recovery***(F_{high}, F_{low});
  end for
Stage 3: Feature Aggregation
  Calculate contribution weights;
  Perform weighted fusion to produce F_{fused};

IV. Experiments

A. Datasets

Through collaboration, UAV swarms can achieve extended perception range, enhanced robustness, and improved flexibility, leading to better performance in various tasks under complex environments. However, there is a lack of available datasets for multidrone collaboration that supports diverse

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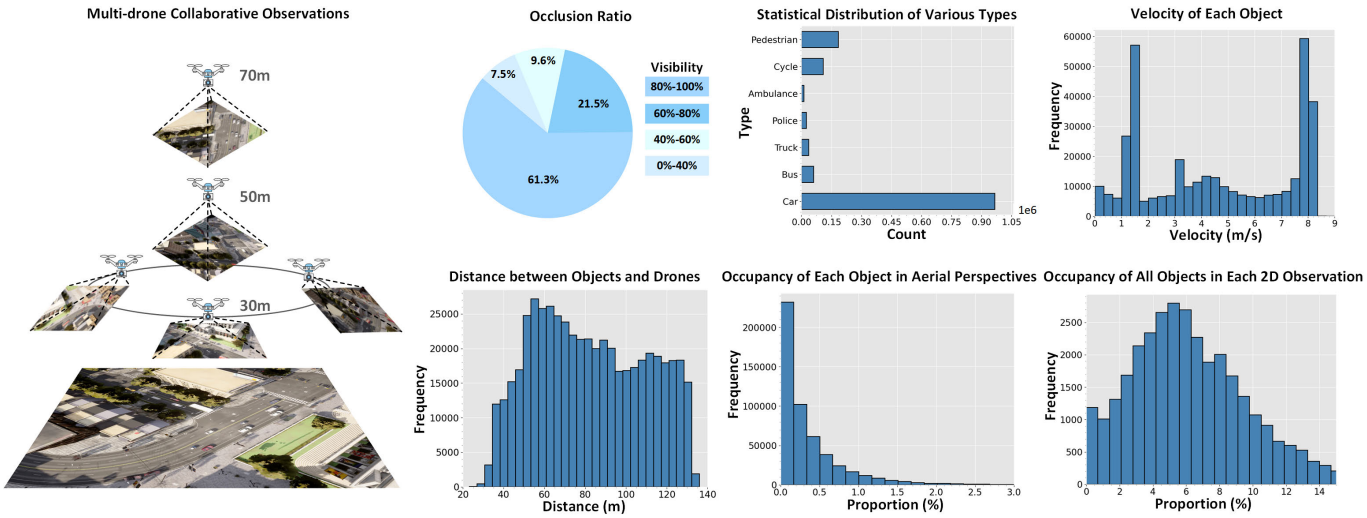


Fig. 5. Illustration on the far left showcases the layout of five drones and the differences in observations collected at various heights and perspectives. The remaining statistical charts reflect specific metrics of the AirCo-MultiTasks dataset, including the occlusion within a single view, the number of various object types, the velocity of each object, the distribution of distances between objects and drones, and the proportion of objects within observations, respectively.

downstream tasks. To provide a more comprehensive and challenging benchmark for the multidrone perception community, we introduce AirCo-MultiTasks, a simulation dataset based on CARLA [49] that supports multidrone collaborative 3-D object detection, BEV semantic segmentation, and target trajectory prediction. The dataset is described in detail as follows.

1) Sensor Setup and Data Collection: Each UAV is equipped with an RGB optical camera featuring a 70° field of view (FOV) and a resolution of 1600 × 900 pixels. A total of five collaborative drones are deployed with varying pitch angles and altitudes to monitor traffic flow at intersections. As illustrated in the left part of Fig. 5, three UAVs are positioned at a height of 30 m, distributed at three corners to form an approximate equilateral triangle. A fourth UAV is placed 50 m above the center of this triangle, while a fifth UAV is positioned at 70 m. This configuration effectively covers an area of approximately 150 × 150 m and aims to explore the advantages of both complementary information from multiple perspectives and out-of-range observations provided by altitude variations. In addition, they capture images at a frequency of 2 Hz to support the temporal prediction task. The AirCo-MultiTasks dataset includes a total of 38 000 synchronous images, divided into 128 training scenes and 32 validation scenes.

2) Data Annotations: Each frame is meticulously annotated with 3-D labels, which include location, rotation in the global coordinate system, size (length, width, and height), and visibility. In addition, the dataset comprises three major object categories: vehicles, cycles, and pedestrians. The annotation format and structure of the dataset are similar to those of the widely used autonomous driving dataset, nuScenes [50], facilitating direct compatibility with the well-established nuScenes-devkit. This toolkit allows for the transformation of 3-D annotations into 2-D labels, BEV segmentation maps, and trajectories, supporting a range of downstream tasks. The annotations required for various tasks in this article are intuitively visualized in Fig. 6.

3) Generation of Traffic Scenes: The CARLA [49] simulation platform generates road layouts, dynamic targets, and traffic flow. Vehicles, bicycles, and pedestrians are controlled via official scripts that adhere to traffic regulations, such as vehicles yielding to pedestrians and cyclists. In addition, to increase the complexity of the scenes, the weather and traffic flow are randomly initialized for each scenario. For scene settings, we select three high-quality urban maps, including CARLA's Town 3, Town 5, and Town 10, and designate 33 data collection locations near intersections for comprehensive observation of traffic flow from multiple perspectives. The chosen areas are similar to realistic urban environments with tall buildings and roadside trees that may cause observation occlusions. These challenges can be effectively solved through multidrone collaboration.

4) Dataset Characteristics: Regarding occlusion issues mentioned previously, the AirCo-MultiTasks dataset includes a substantial number of occluded objects, categorized into four visibility levels. Approximately 10% of the objects have a visibility of less than 40%. In terms of observation distances, most items fall within the range of 60–120 m, which is considered long-distance observation, leading to small objects in aerial perspectives. Specifically, almost each object occupies less than 2% of the FOV, and the total proportion of objects within the FOV is less than 15%. Consequently, this dataset exhibits significant characteristics of remote sensing, with a large proportion of background content, small observed objects, and a low overall proportion of objects in observations. Given the challenges of tiny objects from aerial observations, this study mainly focuses on the vehicle category, which includes various subcategories to enhance the robustness of identifying different types of vehicles. These subcategories are initialized and configured via CARLA scripts. The specific statistics of the AirCo-MultiTasks dataset are provided in the charts shown in Fig. 5.

5) Comparison With Existing Datasets: In remote sensing, several datasets focus on multisatellite and multi-UAV

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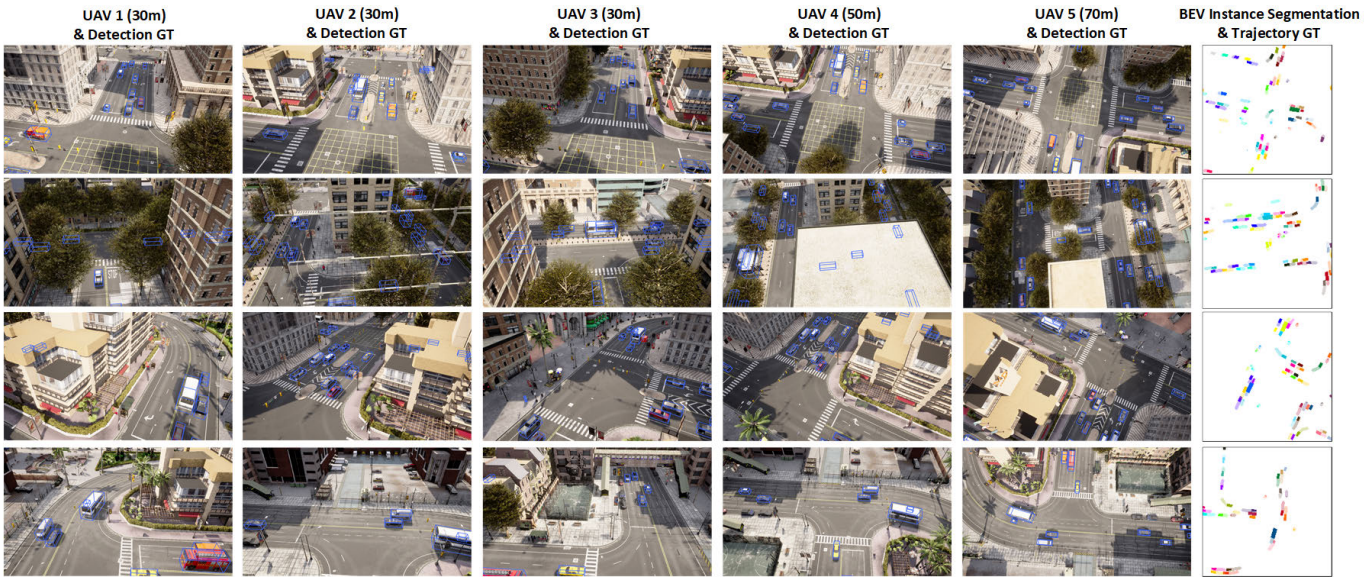


Fig. 6. Visualization of our AirCo-MultiTasks dataset along with the detailed annotations for collaborative 3-D object detection, BEV instance segmentation, and trajectory prediction tasks.

collaboration. For example, the DFC23 dataset [51] enhances segmentation accuracy through the collaboration of Gaofen-2 and SuperView-1 satellites over the same scene. The University-1652 dataset [52] facilitates cross-view geo-localization tasks by leveraging viewpoint-invariant features between UAV and satellite perspectives. The VisDrone series includes datasets such as MDOT, which collects data for joint single-object [53] tracking using two or three UAVs, and multi-drone multi-target tracking (MDMT) [54], which focuses on dual-UAV multiobject tracking. In addition, the AirSim-MAP [35] simulated dataset assists in restoring blurred perspective information through multi-UAV cooperation. However, these existing datasets typically lack camera parameters and position coordinates, which are essential for accurately transforming and fusing multiview information. Moreover, their tasks are relatively singular, limiting their utility for research on foundational models capable of handling multiple tasks. In contrast, our proposed AirCo-MultiTasks dataset addresses these limitations. It includes observations from multiple UAVs at varying altitudes and perspectives, along with corresponding position coordinates, camera parameters, and detailed annotations. This dataset supports a wide range of tasks, including detection, segmentation, and prediction, making it well-suited for multiplatform and multitask research.

B. Evaluation Metrics

1) 3-D Object Detection: The evaluation metrics for 3-D object detection include mean average precision (mAP), mean average translation error (mATE), mean absolute scale error (mASE), and mean absolute orientation error (mAOE). These metrics are defined as follows.

1) *mAP*: This metric measures the average precision (AP) of detected objects across various match thresholds. It uses the AP metric, which defines a match by thresholding the 2-D center distance on the ground plane instead of "intersection over union." AP is calculated as the normalized area under the precision–recall curve, considering only points where recall and precision are both above 10%. This approach helps to minimize the impact of noise often found in low precision and recall regions. Then, an average is computed over matching thresholds $D = \{0.5, 1, 2, 4\}$ m

$$\text{mAP} = \frac{1}{|D|} \sum_{d \in D} AP_d \tag{12}$$

where AP_d is the average precision at threshold d.

2) *mATE*: This metric evaluates the average translation error of the detection bounding box centers. It measures the accuracy of the model in detecting the position of the objects. The specific calculation is as follows:

$$\text{mATE} = \frac{1}{N} \sum_{i=1}^{N} \|c_i - \hat{c}_i\| \tag{13}$$

where c_i and $\hat{c}_i$ are the ground truth and predicted centers of the ith bounding box, respectively, and N is the total number of detected objects.

3) *mASE*: This metric measures the average scale error of the detection bounding boxes. It evaluates the accuracy of the model in estimating the size of the objects. The specific calculation is as follows:

$$\text{mASE} = \frac{1}{N} \sum_{i=1}^{N} \left|s_i - \hat{s}_i\right| \tag{14}$$

where s_i and $\hat{s}_i$ are the ground truth and predicted scales (e.g., width, height, and length) of the ith bounding box, respectively.

4) *mAOE*: This metric assesses the average orientation error of the detection bounding boxes. It evaluates the accuracy of the model in estimating the orientation of

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the objects. The specific calculation is as follows:

$$\text{mAOE} = \frac{1}{N}\sum_{i=1}^{N}\left|\theta_i - \hat{\theta}_i\right| \tag{15}$$

where θ_i and $\hat{\theta}_i$ are the ground truth and predicted orientations of the ith bounding box, respectively.

2) Instance Segmentation: We adopt four metrics widely used in previous works [55], [56], including intersection-over-union (IoU), segmentation quality (SQ), recognition quality (RQ), and panoptic quality (PQ). These metrics are defined as follows.

1) *IoU*: This metric evaluates the SQ of objects. It is calculated as follows:

$$\text{IoU}\left(\hat{y}^{\text{seg}}, y^{\text{seg}}\right) = \frac{\sum_{h,w}\hat{y}^{\text{seg}} \cdot y^{\text{seg}}}{\sum_{h,w}\hat{y}^{\text{seg}} + y^{\text{seg}} - \hat{y}^{\text{seg}} \cdot y^{\text{seg}}} \tag{16}$$

where $\hat{y}^{\text{seg}}$ and y^{seg} denote the predicted and ground-truth semantic segmentation, respectively.

2) *SQ*: This metric evaluates the SQ of the correctly detected segments. It is the average IoU of the true positives (TPs)

$$\text{SQ}\left(\hat{y}^{\text{inst}}, y^{\text{inst}}\right) = \frac{\sum_{(p,q)\in\text{TP}}\text{IoU}(p,q)}{|\text{TP}|}. \tag{17}$$

3) *RQ*: This metric evaluates the RQ of the predicted segments, considering both TPs and the penalties for false positives (FPs) and false negatives (FNs)

$$\text{RQ}\left(\hat{y}^{\text{inst}}, y^{\text{inst}}\right) = \frac{|\text{TP}|}{|\text{TP}| + \frac{1}{2}|\text{FP}| + \frac{1}{2}|\text{FN}|}. \tag{18}$$

4) *PQ*: This metric considers both the quality of segmentation and the accuracy of instance recognition, capturing both the overlap between predicted and ground-truth segments and the correct classification of instances. The metric is expressed as follows:

$$\text{PQ}\left(\hat{y}^{\text{inst}}, y^{\text{inst}}\right) = \frac{\sum_{(p,q)\in\text{TP}}\text{IoU}(p,q)}{|\text{TP}| + \frac{1}{2}|\text{FP}| + \frac{1}{2}|\text{FN}|}. \tag{19}$$

TP, FP, and FN correspond to true positive, false positive, and false negative, respectively. TP are the correctly predicted segments with an IoU greater than 0.5 that match ground-truth segments. FPs are predicted segments that do not correspond to any ground-truth segments. FNs are ground truth segments that the model fails to predict.

3) Trajectory Prediction: We adopt IoU and Video PQ (VPQ) [56], [57] as the temporal evaluation metrics.

1) *IoU*: For frame-level evaluation, IoU evaluates the SQ of objects at the present and future frames. The specific calculation is as follows:

$$\text{IoU}\left(\hat{y}_t^{\text{seg}}, y_t^{\text{seg}}\right) = \frac{1}{N}\sum_{t=0}^{N-1}\frac{\sum_{h,w}\hat{y}_t^{\text{seg}} \cdot y_t^{\text{seg}}}{\sum_{h,w}\hat{y}_t^{\text{seg}} + y_t^{\text{seg}} - \hat{y}_t^{\text{seg}} \cdot y_t^{\text{seg}}} \tag{20}$$

where N denotes the number of output frames and $\hat{y}_t^{\text{seg}}$ and y_t^{seg} denote the predicted and ground-truth semantic segmentation at timestamp t, respectively.

2) *VPQ*: For video-level evaluation, VPQ measures the quality of the segmentation and ID consistency of the instances through the video, expressed as follows:

$$\text{VPQ}\left(\hat{y}_t^{\text{inst}}, y_t^{\text{inst}}\right) = \sum_{t=0}^{N-1}\frac{\sum_{(p_t,q_t)\in\text{TP}_t}\text{IoU}(p_t,q_t)}{|\text{TP}_t| + \frac{1}{2}|\text{FP}_t| + \frac{1}{2}|\text{FN}_t|} \tag{21}$$

where TP_t, FP_t, and FN_t correspond to true positives, false positives, and false negatives at timestamp t, respectively. This metric effectively reflects the quality of predicted trajectories by considering both future instances' segmentation and identification consistency across multiple frames.

C. Implementation Details

We adhere to the setups introduced in previous studies [57], [58], [59] for various BEV-specific downstream tasks. Initially, the raw images, with a resolution of 900 × 1600 pixels, are scaled and randomly cropped to 256 × 704 pixels. For view transformation, the relative depth estimation range is set from 0 to 1 and discretized into 100 intervals. In contrast, traditional absolute depth estimation methods, for comparison, have a range of 20–140 m, also divided into 100 intervals. For BEV representations, the spatial ranges for the x-, y-, and z-axes are configured as $[-75, 75]$, $[-75, 75]$, and $[0, 10]$ m, respectively. The model performance is evaluated in two perceptual scopes: a 150 × 150 m area (long range) and a 75 × 75 m area (short range), both with a 0.75-m resolution. Specifically, we adopt Swin-Tiny [43] from the family of foundation models as the backbone for each drone due to its lightweight design and low inference latency. For the HLFDC module, the size of the local window in the high-frequency branch and the stride of the average pooling in the low-frequency branch are both set to 4. Regarding downstream tasks, BEVDet [58] is used as the detector for 3-D object detection, while Fiery [59] serves as the segmenter for BEV instance segmentation. For BEV trajectory prediction, PowerBEV [57] is employed as the predictor, where the input consists of three consecutive frames, and the output generates four future frames. Our proposed framework is trained using the AdamW optimizer with an initial learning rate of 2×10^{-4}. The training process is conducted on four NVIDIA A40 GPUs, with a batch size of 8 for 20 epochs.

D. Comparative Baselines for Collaborative Perception

Early collaboration involves the integration of raw observations, where multiview images are combined to generate BEV representations.

Intermediate collaboration focuses on feature-level interactions to produce comprehensive BEV representations. These approaches can be either fully connected or partially connected:

1) The fully connected paradigm shares complete features among all members.
   a) V2VNet [60] performs multiround message exchanges using graph neural networks.

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TABLE I
ANALYSIS OF TRANSMISSION EXPENSE, PARAMETER AMOUNT, AND COMPUTATIONAL COST ACROSS DIFFERENT COLLABORATION STRATEGIES

Models	Average Expense (MB)	Trainable Params (M)	Model Params Size (MB)	Total FLOPs (G)
No Collaboration	-	70.1	281.83	129.87
Early Collaboration	4.12	70.1	281.83	364.73
Late Collaboration	0.61	70.1	281.83	644.48
V2X-ViT	9.77	70.2	282.17	389.84
V2VNet	9.77	70.5	283.47	1205.72
Who2com	2.44	75.5	303.32	375.90
When2com	3.13	75.5	303.32	378.14
Where2comm	4.89	70.1	281.83	364.82
RingMo-Galaxy (Ours)	5.19	70.2	282.01	383.70

   b) vehicle-to-everything (V2X)-ViT [61] explores correlations among heterogeneous collaborators through transformer blocks.

2) The partially connected paradigm restricts interactions to certain members or regions.
   a) Who2com [34] introduces a handshake mechanism to select collaborative partners.
   b) When2com [35] determines the optimal timing for initiating collaboration.
   c) Where2comm [11] uses the detection head to direct interactions toward specific regions.

Late collaboration combines individual prediction results from multiple platforms to form a final decision.

Table I presents the transmission expense, parameter amount, and computational cost for each comparative baseline and our RingMo-Galaxy framework. To provide a comprehensive comparison, we use a complex prediction task that represents the upper limit of parameter count and computational cost among these tasks. The transmission expense is measured in megabytes (MB) to clearly illustrate the data volume exchanged during collaborative interactions. In the specific implementation, shared encoders and decoders are utilized for feature extraction and task interpretation. Variations in model parameters and computational costs arise from the design of different collaborative strategies.

E. Quantitative Evaluation

1) Benchmark Comparison in 3-D Object Detection: Table II demonstrates that our proposed approach, RingMo-Galaxy, achieves the highest performance in terms of mAP and mATE among all collaborative perception methods, even much better than the early collaboration without incorporating generalized feature mapping. This superior performance can be attributed to our designed GBG module, which facilitates more accurate object localization and yields outstanding results in position-related metrics. Although our RingMo-Galaxy exhibits slightly lower performance in mASE and mAOE, it still substantially reduces these errors compared with most collaborative approaches. The slightly lower performance in mASE and mAOE may be due to the inherent challenges in accurately estimating object scale and orientation under multiview conditions, which are not the primary focus of our current model design. In addition, RingMo-Galaxy significantly outperforms the no-collaboration baseline, achieving a 202% increase in mAP for the short range and a 254% increase in mAP for the long range, indicating the effectiveness of collaboration. In contrast to fully connected baselines of intermediate collaboration, RingMo-Galaxy enhances object detection accuracy and recall by 2%–4% and reduces errors in localization, scale, and orientation by 6%–30%, while requiring only approximately half the transmission ratio. When compared with the partially connected baselines of intermediate collaboration, RingMo-Galaxy surpasses the previous state-of-the-art method, Where2com, with a 4.4% increase in mAP, a 6% decrease in mATE, a 10% decrease in mASE, and a 9.5% decrease in mAOE in the long-range setting. The key difference between our RingMo-Galaxy and Where2comm is that Where2comm utilizes confidence matrices to select spatial information for sparse feature transmission, which results in spatial discontinuity in the transmitted features. Although some methods have attempted to smooth features via interpolation, this nonparametric approach may generate spurious target information by interpolating between target and empty features. In contrast, our proposed method considers both spatial and channel dimensions. RingMo-Galaxy compresses the low-frequency components of the global spatial structure while preserving the high-frequency components of local fine details, then appropriately compresses both in the channel dimension. This dual-branch design not only reduces transmission costs but also ensures the spatial continuity of transmitted collaborative features by leveraging the complementary characteristics of high and low frequency components.

2) Benchmark Comparison in BEV Instance Segmentation: As presented in Table III, our proposed RingMo-Galaxy demonstrates the top performance in instance segmentation across all evaluation metrics among collaborative approaches. Compared with the second-best early collaboration approach, which processes multiview raw observations in a centralized manner, RingMo-Galaxy improves the IoU by 2.8%–3.1% and PQ by 2.4%–5.1% through feature-level interactions, facilitated by our designed GBG and HLFDC modules. The well-known collaborative detection baseline, Where2comm, which utilizes detection heads as confidence guides for spatially sparse transmission, is not well-suited for collaborative instance segmentation. The performance gap in the comprehensive metric, PQ, between Where2comm and both the fully connected baselines of intermediate collaboration and our proposed RingMo-Galaxy exceeds 30%. Furthermore, Who2com, which selects only one perception helper at a time, gathers less complementary information than multiple collaborators, resulting in limited improvement over no collaboration, as expected. Notably, When2comm shows hardly any performance improvement. Because it determines collaboration opportunities based on mutual similarity, which is ineffective due to the significant variation in multiview features, ultimately converging into a dilemma where inter-platform collaboration is unnecessary. Unlike the intricate task of 3-D object detection, which considers orientation,

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TABLE II
COMPARISON OF COLLABORATION BASELINES FOR 3-D OBJECT DETECTION

Methods		mAP ↑		mATE ↓		mASE ↓		mAOE ↓	
		Short	Long	Short	Long	Short	Long	Short	Long
No Collaboration		0.219	0.125	0.570	0.631	0.195	0.204	0.182	0.190
Early Collaboration		0.641	0.437	0.382	0.430	**0.160**	**0.169**	**0.065**	**0.083**
Late Collaboration		0.506	0.303	0.416	0.423	0.212	0.213	0.101	0.103
Intermediate Collaboration (Fully Connected)	V2X-ViT [61]	0.638	0.434	0.410	0.457	0.240	0.208	0.094	0.116
	V2VNet [60]	0.631	0.433	0.403	0.453	0.184	0.190	0.088	0.108
Intermediate Collaboration (Partially Connected)	Who2com [34]	0.461	0.291	0.522	0.580	0.188	0.196	0.116	0.135
	When2com [35]	0.137	0.063	0.495	0.513	0.185	0.189	0.222	0.237
	Where2comm [11]	0.634	0.424	0.398	0.447	0.189	0.196	0.118	0.136
	RingMo-Galaxy (Ours)	**0.662**	**0.443**	**0.374**	**0.413**	0.170	0.176	0.109	0.123

instance segmentation focuses on recognizing and segmenting objects to reflect recall and localization accuracy. Attributed to our GBG module improving feature mapping from 2-D to BEV, our RingMo-Galaxy surpasses prevalent fully connected collaboration approaches by 3%–4.5% in IoU and 1.8%–5% in PQ. In contrast to previous sparse collaborative perception methods that reduce transmission costs at the expense of performance, our method outperforms these baselines by 35%–75% in IoU and 42%–63% in PQ.

3) Benchmark Comparison in Trajectory Prediction: As presented in Table IV, it is challenging to predict objects' future trajectories solely based on single-view observation, and the performance of both future instance segmentation and motion prediction significantly improves with the help of any collaborative methods. Distinct from observation patterns of single-frame instance segmentation, previous collaborative approaches, such as early collaboration, V2XViT, and V2VNet, exhibit comparative performance with our method in the short range. This suggests that temporal information can mitigate the estimation error of feature mapping. However, the gap between our method and previous collaborative methods becomes evident in the long-range setting. Our RingMo-Galaxy outperforms early collaboration by 2.5% in IoU and 2% in VPQ. Compared with fully connected collaborative approaches, our method surpasses V2X-ViT by 3.8% in IoU and 5.7% in VPQ and V2VNet by 17.8% in IoU and 12.8% in VPQ. Notably, even with the same transmission ratio as the partially connected baseline Where2comm, our method significantly outperforms it by 10.8% in IoU and by 42.4% in VPQ, highlighting the effectiveness of our HLDFC module. In addition, similar to the results observed in collaborative instance segmentation, the performance gap between traditional partially connected collaborative methods (Who2com, When2com, and Where2comm) and both fully connected baselines and our proposed RingMo-Galaxy remains significant in collaborative trajectory prediction. These partially connected methods attempt to balance transmission cost and performance improvement, but aside from detection tasks, they even perform worse than the late collaboration approach, which incurs less overhead.

F. Ablation Studies

1) Effectiveness of the Proposed Modules: Our collaborative framework, RingMo-Galaxy, introduces two innovative components: the GBG and HLFDC modules. We evaluate these modules based on their ability to enhance downstream tasks in the long-range setting, including collaborative 3-D object detection, BEV instance segmentation, and trajectory prediction, as well as their ability to optimize the performance-transmission tradeoff, as depicted in Table V. The GBG-only variant significantly enhances 3-D object detection accuracy, a roughly 4% increase in mAP and a 7.2% reduction in mATE. This is primarily attributable to the more accurate BEV representations guided by the generalized feature mapping. However, this also leads to the recall of harder-to-detect objects, particularly those with challenging orientations and scales. While these cases are successfully recalled, they introduce higher orientation (mAOE) and scale (mASE) errors. As a result, despite the mAP improvement from better detection and localization, the overall errors in size and orientation increase. For instance segmentation, the GBG-only variant improves segmentation and RQ by 2.1% and 1.4%, respectively. As to trajectory prediction, this variant also shows improvements of 6.7% in IoU and 6.1% in VPQ. The evaluation metrics for these tasks are highly related to position precision, thereby demonstrating the effectiveness of our designed GBG module. On the other hand, the HLFDC-only variant reduces transmission costs by 46.9%, with only a marginal performance decrease of less than 0.5% in mAP, IoU, and VPQ, relative to the baseline without any modules. This indicates the validity of the HLFDC, albeit with minor performance tradeoffs. Overall, our RingMo-Galaxy, integrated with both GBG and HLFDC modules, achieves a balance between collaborative perception and transmission efficiency.

2) Effects of the High and Low Branches in HLFDC Module: Following the verification of the GBG module's effectiveness, we further investigate the influence of high- and low-frequency branches within the HLFDC module on various downstream tasks. The relevant results are presented in Table VI. As expected, transmitting the original features offers

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TABLE III
COMPARISON OF COLLABORATION BASELINES FOR BEV INSTANCE SEGMENTATION

Models		IoU (%) ↑		PQ (%) ↑		SQ (%) ↑		RQ (%) ↑	
		Short	Long	Short	Long	Short	Long	Short	Long
No Collaboration		37.02	18.74	31.59	17.06	72.14	71.60	43.78	23.82
Early Collaboration		62.15	29.66	49.23	26.67	74.13	73.63	66.41	36.22
Late Collaboration		53.66	26.10	31.12	17.41	67.65	67.54	46.01	25.78
Intermediate Collaboration (Fully Connected)	V2X-ViT [61]	62.10	29.84	49.74	26.49	73.87	73.19	67.33	36.20
	V2VNet [60]	61.68	30.25	48.60	26.83	73.47	72.98	66.15	36.77
Intermediate Collaboration (Partially Connected)	Who2com [34]	47.10	21.73	35.89	18.76	70.54	70.41	50.88	26.65
	When2com [35]	36.45	18.40	31.17	16.85	71.34	70.73	43.69	23.83
	Where2comm [11]	51.63	25.69	35.60	19.68	67.75	67.50	52.54	29.15
	RingMo-Galaxy (Ours)	**63.92**	**31.17**	**51.04**	**27.30**	**75.47**	**74.76**	**67.63**	**36.51**

TABLE IV
COMPARISON OF COLLABORATION BASELINES FOR TRAJECTORY PREDICTION

Models		IoU (%) ↑		VPQ (%) ↑	
		Short	Long	Short	Long
No Collaboration		21.00	11.21	19.66	7.26
Early Collaboration		52.82	34.17	39.84	26.01
Late Collaboration		44.89	28.32	28.42	17.94
Intermediate Collaboration (Fully Connected)	V2X-ViT [61]	52.49	33.58	37.87	24.54
	V2VNet [60]	52.31	32.91	39.68	26.03
Intermediate Collaboration (Partially Connected)	Who2com [34]	42.94	26.98	30.60	19.80
	When2com [35]	22.18	11.48	24.52	15.16
	Where2comm [11]	43.10	27.93	25.45	16.76
	RingMo-Galaxy (Ours)	**53.46**	**34.51**	**41.75**	**26.42**

the best performance for joint interpretation tasks, as it maximally retains information from the observations. However, this manner also incurs the highest transmission overhead. When only with the high-frequency branch, the transmission cost is reduced to 50% of the original, while the low-frequency branch alone reduces it further to just 3.1%. For object recognition and segmentation, merely providing the high-frequency component significantly outperforms low-frequency transmission due to its richer edge details and higher information content. Interestingly, for sequential tasks, such as trajectory prediction, transmitting lightweight low-frequency information leads to notable performance gains, even outperforming high-frequency transmission. This may be because the temporal structure compensates for the lack of single-frame details, and the coarse scene layout and approximate object position provided by the low-frequency components are more valuable for trajectory prediction. When both high- and low-frequency branches are combined, performance improves significantly compared to using either branch alone. With just 53.1% of the original transmission cost, this dual-branch approach delivers comparable results, with performance degradation limited to 2.6%, 1.2%, and 1.4% across the three downstream tasks. This further highlights the effectiveness of our high–low frequency decoupling approaches in balancing performance and transmission efficiency.

3) Effect of Window Size in HLFDC Module: Table VII lists the comparison of the performance metrics for different window sizes implemented in the HLFDC module. The window size determines the contextual range in each local window for high-frequency detail extraction and the compression ratio for low-frequency information preservation. Smaller windows are more sensitive to capturing high-frequency local feature variations but may be susceptible to noise, resulting in considerable transmission costs for collaborative interactions. Conversely, larger windows encompass a broader range of textual information within the local windows. However, they may potentially lose significant information via low-frequency global attention processing due to excessive information compression. After investigating the effects of various window sizes across different downstream tasks, we select a 4×4 window size in the HLFDC module for its optimal balance between performance and transmission efficiency.

4) Number of the Participants in the Collaborative Framework: We conduct the complementary experiments to evaluate the performance variation as the number of platforms increases within the long-range setting, as shown in Table VIII. For all three downstream tasks, we observe a general trend of performance improvement with the addition of more drones, as expected. Conversely, a reduction in the number of drones demonstrates the impact of potential drone failures on performance metrics. It is worth noting that the performance gains from adding the fourth drone (from 3 to 4) are less significant compared with the improvements observed when adding the third or fifth drone. This can be attributed to the substantial overlap in coverage between the three drones (positioned at a 30m inclined view) and the fourth drone (positioned at a 40-m top–down view). Consequently, the performance enhancement brought by the fourth drone is relatively diminished due to this redundancy in coverage.

G. Visualization

1) Collaborative 3-D Object Detection: Fig. 7 illustrates that collaborative perception significantly enhances the

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TABLE V

ABLATION STUDY OF THE PROPOSED MODULES. "RATIO" IN THE TABLE REFERS TO THE RELATIVE TRANSMISSION COST OF EACH VARIANT COMPARED TO THE COMPLETE FEATURE TRANSMISSION DURING COLLABORATIVE INTERACTIONS

GBG	HLFDC	3D Object Detection				BEV Instance Segmentation				Trajectory Prediction		Ratio
		mAP ↑	mATE ↓	mASE ↓	mAOE ↓	IoU (%) ↑	PQ (%) ↑	SQ (%) ↑	RQ (%) ↑	IoU (%) ↑	VPQ (%) ↑	
×	×	0.437	0.430	0.169	0.083	29.66	26.67	73.63	36.22	34.77	28.70	1.000
✓	×	0.455	0.399	0.172	0.135	31.24	27.65	75.24	36.74	37.09	30.46	1.000
×	✓	0.433	0.443	0.189	0.105	29.64	26.05	72.81	35.77	34.59	28.35	0.531
✓	✓	0.443	0.413	0.176	0.123	31.17	27.30	74.76	36.51	36.71	39.58	0.531

TABLE VI

ABLATION STUDY ON THE CONTRIBUTIONS OF HIGH-FREQUENCY AND LOW-FREQUENCY BRANCHES WITHIN THE HLFDC MODULE

High	Low	3D Object Detection				BEV Instance Segmentation				Trajectory Prediction		Ratio
		mAP ↑	mATE ↓	mASE ↓	mAOE ↓	IoU (%) ↑	PQ (%) ↑	SQ (%) ↑	RQ (%) ↑	IoU (%) ↑	VPQ (%) ↑	
×	×	0.455	0.399	0.172	0.135	31.24	27.65	75.24	36.74	37.09	30.46	1.000
✓	×	0.402	0.435	0.173	0.093	29.33	25.47	71.50	35.62	32.39	22.71	0.500
×	✓	0.309	0.628	0.289	0.155	26.64	17.19	65.24	26.24	35.12	24.35	0.031
✓	✓	0.443	0.413	0.176	0.123	31.17	27.30	74.76	36.51	36.71	29.58	0.531

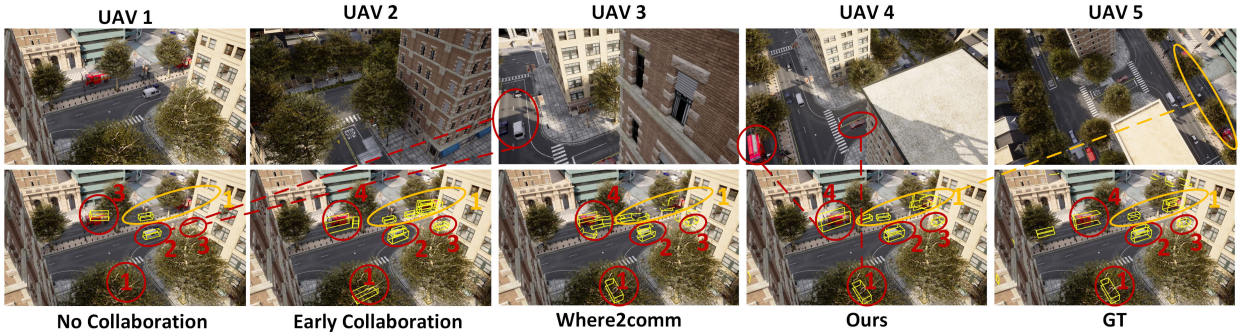


Fig. 7. Visualization of collaborative 3-D object detection. Red circles highlight regions where collaborative methods have successfully corrected errors resulting from single-platform perception, while yellow circles highlight remaining failure cases.

TABLE VII

COMPARISON OF PERFORMANCE METRICS FOR DIFFERENT WINDOW SIZES IMPLEMENTED IN THE HLFDC MODULE. FOR SIMPLIFICATION, WE USE 3D-DET-MAP TO REPRESENT MAP FOR 3-D OBJECT DETECTION, BEV-INST-SEG-IOU, AND TRAJ-PRED-IOU TO REPRESENT IOU FOR BEV INSTANCE SEGMENTATION AND TRAJECTORY PREDICTION, RESPECTIVELY

Window Size	1	2	4	8	16	32
3D-Det-mAP	0.458	0.447	0.443	0.432	0.426	0.421
BEV-Inst-Seg-IoU	31.21	30.79	31.17	30.58	30.23	30.05
Traj-Pred-IoU	34.72	34.45	34.51	34.14	33.90	33.73
Ratio	1.000	0.625	0.531	0.508	0.502	0.500

detection of occluded and out-of-range objects through inter-platform interactions, compared with no collaboration. For instance, UAV 1 struggles to recognize certain objects due to severe occlusion caused by dense roadside trees in its single-view observation. However, collaborative approaches address this limitation by leveraging complementary information from other platforms. As shown in red circles 1 and 2, objects occluded by surrounding vehicles and buildings in UAV 1's view are successfully recalled with the assistance of UAV 3.

Similarly, the target obscured by trees, highlighted in red circle 3, is effectively detected with the help of UAV 4. Notably, the bus blocked by trees is localized with greater accuracy, as indicated by red circle 4. Yellow circle 1 represents a challenging region with heavily occluded objects that are almost invisible in the views of UAVs 1–4 and only partially captured by UAV 5. Due to UAV 5's higher altitude, the targets appear smaller with limited visual details. After 2-D-to-BEV projection and feature-level interactions, most collaborative methods suffer from false detections and multiple redundant bounding boxes in this region. Although our method is not entirely immune to these issues, it significantly reduces such errors compared to other approaches, further demonstrating its superior performance.

2) Collaborative BEV Instance Segmentation: The task of BEV instance segmentation focuses on segmenting the

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TABLE VIII
COLLABORATIVE PERFORMANCE ACROSS VARIOUS DOWNSTREAM TASKS WITH RESPECT TO THE NUMBER OF PARTICIPATING PLATFORMS

Num	3D Object Detection				BEV Instance Segmentation				Trajectory Prediction	
	mAP ↑	mATE ↓	mASE ↓	mAOE ↓	IoU (%) ↑	PQ (%) ↑	SQ (%) ↑	RQ (%) ↑	IoU (%) ↑	VPQ (%) ↑
1	0.105	0.732	0.279	0.218	12.18	7.65	63.52	12.04	11.94	9.50
2	0.177	0.758	0.285	0.099	20.56	14.97	67.10	22.31	23.67	18.35
3	0.363	0.487	0.191	0.092	22.33	17.39	69.75	20.63	28.04	22.91
4	0.397	0.431	0.192	0.094	27.55	23.02	71.20	32.33	32.98	25.79
5	0.443	0.413	0.176	0.123	31.17	27.30	74.76	36.51	34.51	26.42

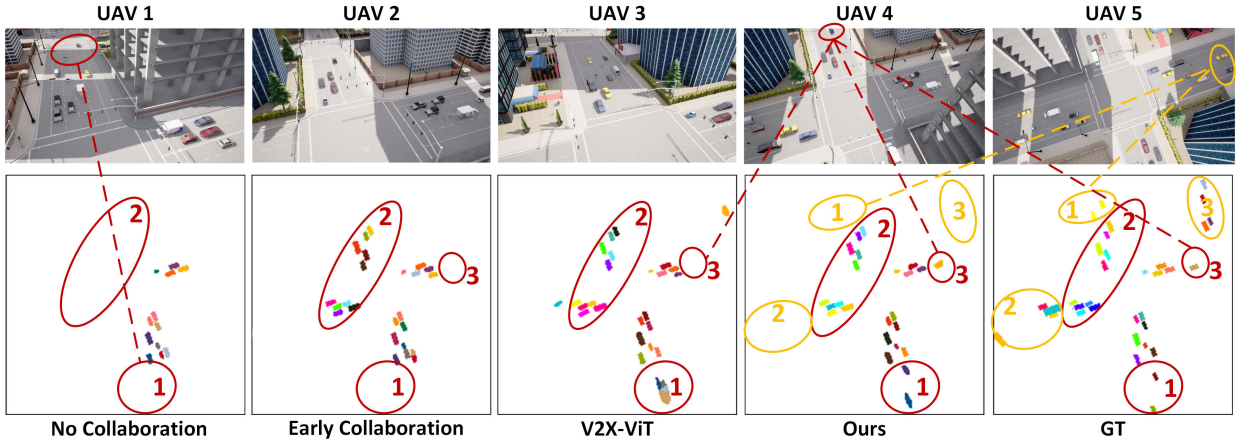


Fig. 8. Visualization of collaborative instance segmentation. Each instance is allocated a distinct color. Red circles highlight regions where collaborative methods have successfully corrected errors resulting from single-platform perception, while yellow circles highlight remaining failure cases.

BEV representation into small object bricks, where the accuracy heavily depends on precise 2-D-to-BEV mapping. For visualization, we select early collaboration and V2X–ViT as baselines due to their competitive performance, as presented in Table III. As illustrated in Fig. 8, the single-view observation without collaboration can only identify large objects close to the camera, failing to identify distant and smaller objects. For instance, the vehicles located at the top of UAV 1's FOV are not identified, as indicated by red circle 1 in the first column of the prediction results. In contrast to the other baselines, our proposed RingMo-Galaxy successfully recalls these objects with higher positional accuracy, demonstrating its superiority in accurately segmenting distant targets. For the single-view platform, UAV 1, a significant amount of out-of-range objects at intersections are missing. Collaborative methods can alleviate this issue by leveraging inter-platform complementary information, as shown in the visualizations of columns 2–4, particularly within the red circle 2. While early collaboration centralizes the processing of raw observations, it performs less effectively than our approach, which requires less transmission overhead through compressed feature-level interactions of high-low frequency dual branches. In addition, for the targets highlighted by red circle 3, although they are recalled by both methods, early collaboration exhibits noticeable positional deviations compared to ours. This indicates that precise feature mapping is more significant than transmitting a larger quantity of features, reaffirming the superiority of our RingMo-Galaxy in BEV instance segmentation. Nevertheless, some failure cases remain, as highlighted by the yellow circles in the visualization. Yellow circle 1 highlights instances where our method fails to recall objects typically positioned at the edges of aerial observations. Meanwhile, the targets marked by yellow circles 2 and 3, although included in the ground truth, are not recalled. This is understandable, as these targets fall within the limited-visibility areas of all platforms, making their detection inherently impossible.

3) Collaborative Trajectory Prediction: The trajectory prediction heavily depends on interframe continuity. Previous sparse collaborative approaches, like Where2comm, convey spatially discrete features, potentially disrupting this continuity. In contrast, our method preserves the overall structure of intermediate BEV features and extracts critical information through high- and low-frequency decoupling and compression. In addition, our framework also benefits from a more accurate generalized feature mapping module. As a result, our method achieves performance comparable with or even better than early collaboration and fully connected intermediate collaboration paradigms. By integrating our specially designed GfBG and HLFDC modules, RingMo-Galaxy accurately forecasts the trajectories of multiple objects in challenging intersections and achieves more precise segmentation and prediction results than other well-known baselines. These findings are consistent with our quantitative evaluations. As shown in Fig. 9, by incorporating supplementary information from other UAVs, these collaborative methods not only recall additional objects but also improve trajectory predictions. For instance, red circle 1 highlights how interplatform collaboration successfully recalls objects beyond UAV 1's perspective, while

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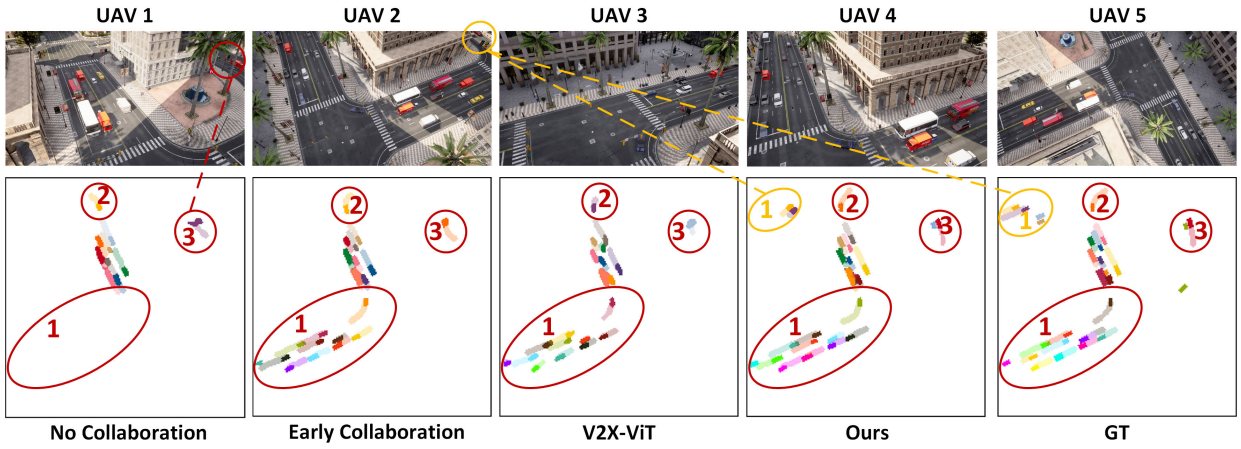


Fig. 9. Visualization of collaborative trajectory prediction. Each instance is allocated a distinct color, and its predicted trajectory is represented with the same color and slight transparency. Red circles highlight regions where collaborative methods have successfully corrected errors resulting from single-platform perception, while yellow circles highlight remaining failure cases.

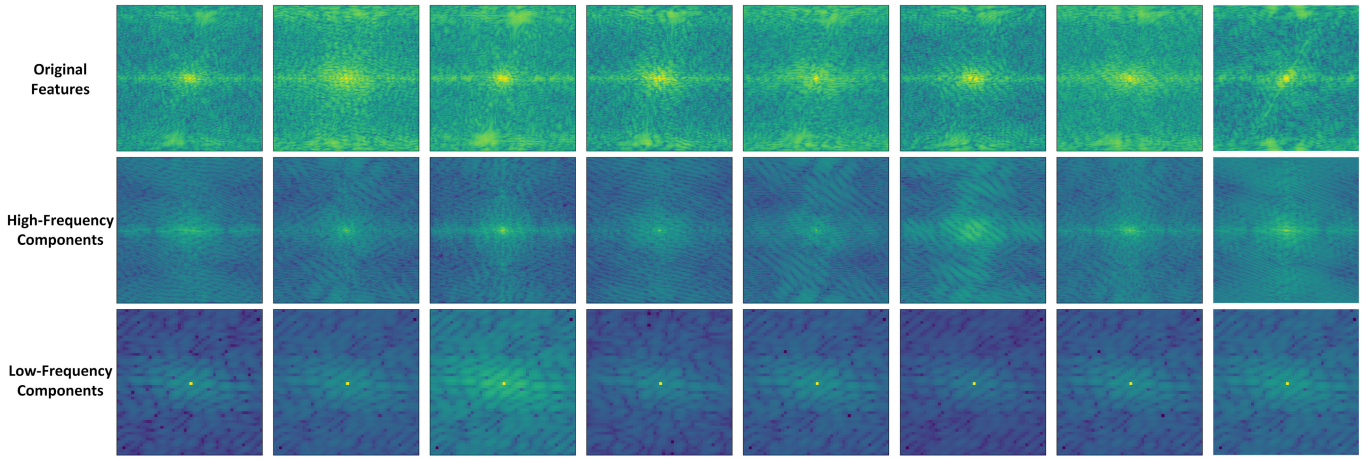


Fig. 10. Visualization of high- and low-frequency feature decoupling in the HLFDC module through FFT in the frequency domain. Lighter colors indicate greater magnitude, with regions nearer the center representing lower frequencies.

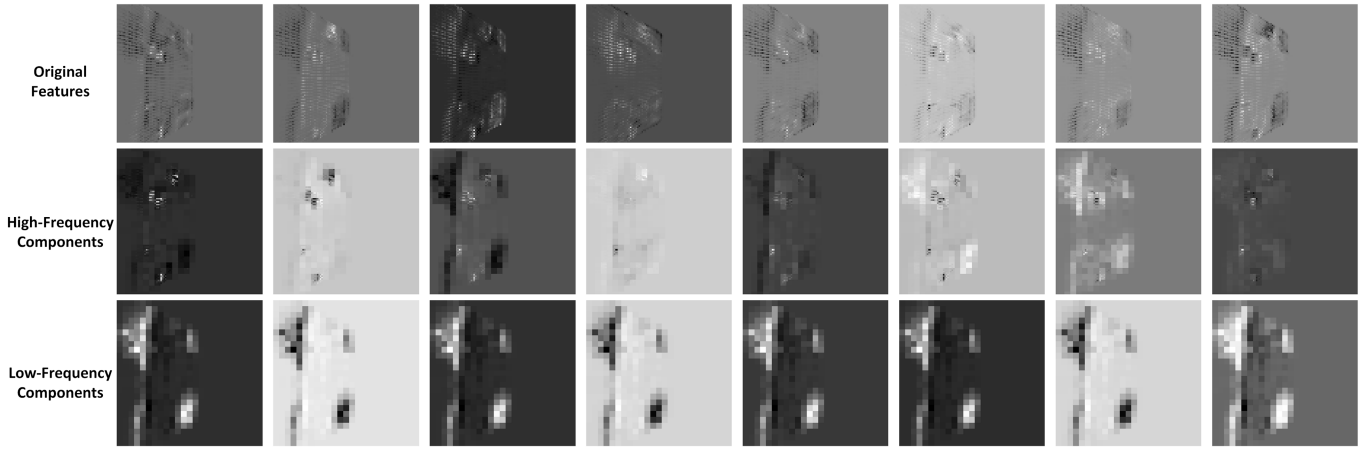


Fig. 11. Visualization of high- and low-frequency feature decoupling in the HLFDC module from the perspective of the spatial domain. Areas with lighter colors indicate greater emphasis.

red circle 2 demonstrates the correction of erroneous trajectory predictions. Red circle 3 highlights a challenging case located at the edge of UAV 1's observation range. While other methods fail due to traditional feature mapping limitations, leading to partial object omissions, our approach, equipped with a generalized feature mapping module, successfully recalls the complete object information. However, as highlighted by yellow circle 1, some failure cases remain. Specifically, two

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objects are lost due to their location in the limited-visibility areas of all platforms, which is understandable. In addition, the model erroneously predicts motion trajectories for two static targets. This may result from temporal projection inconsistencies, where the recalled targets are misinterpreted as moving objects, leading to incorrect trajectory predictions.

4) Analysis of High- and Low-Frequency Decoupling: We conduct a qualitative analysis to intuitively demonstrate the high- and low-frequency information decoupled by the HLFDC module, as shown in Figs. 10 and 11. For conciseness, we visualize specific channels of the feature maps at fixed intervals, extracting eight channels from the identical features. In Fig. 10, we visualize the magnitudes of frequency components' magnitudes by applying a fast Fourier transform (FFT) to the original features and decoupled features from the HLFDC module, respectively. Compared with the frequency domain distribution of the original features, the high-frequency components in the second row exhibit a brighter outer region and a darker central peak. Conversely, the low-frequency components' energy in the third row is primarily gathered near the center. In addition, in Fig. 11, we visualize these features from a spatial domain perspective. The high-frequency components in the second row appear to focus on local details, while the low-frequency components in the third row capture broader structural features. The contrast between the two rows visually demonstrates the complementary characteristics of the high- and low-frequency components. These visualizations indicate that our designed high-frequency branch captures more local fine details, while the low-frequency branch focuses mainly on smooth structural features. This outcome strongly aligns with the intended function of the HLFDC module.

V. Conclusion and Discussion

We propose a novel distributed remote sensing foundation model, RingMo-Galaxy. The core idea is to utilize the geometric priors of oblique perspectives to generalize and map features from observations across different platforms into a unified feature space, facilitating subsequent multitask perception. Simultaneously, we decouple high-frequency and low-frequency features and perform selective information compression to mitigate the loss of critical information during compression. Comprehensive quantitative and qualitative experiments, covering multialtitude, multiperspective inputs, and multi-task outputs, demonstrate that RingMo-Galaxy achieves an excellent balance between perception performance and transmission costs in multiplatform collaborative online inference.

The current work primarily focuses on a simulated setting, an idealized scenario, for multidrone collaboration across various downstream tasks. The future research will extend these efforts to real-world environments to enhance practical applicability, addressing realistic challenges, such as sensor noise, latency, and communication frequency.

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